

Lecture X

HTTP and CDNs

CS 168, Fall 2024 @ UC Berkeley

Slides credit: Sylvia Ratnasamy, Rob Shakir, Peyrin Kao, Iuniana Oprescu

HTTP Specification

Lecture X, CS 168, Fall 2024

HTTP

- **Protocol Specification**
- Examples
- Speeding Up HTTP

Content Delivery Networks

- Deployment
- Directing Clients to Caches

Newer HTTP Versions

Brief History of HTTP

Development initiated by Tim Berners-Lee at CERN in 1989.

- 1991: Initial specification, HTTP/0.9, drafted.
- 1996: Standardized as HTTP/1.0.
- 1997: Updated to HTTP/1.1.
 - We'll use this version unless otherwise specified.

Driven by a need to share information between scientists.

- Needed a mechanism to transfer *hypertext* pages, with links to other pages.
- Resulting protocol: **HyperText Transfer Protocol**.

You can still view [the first website ever made](#).

HTTP is a **client-server** protocol.

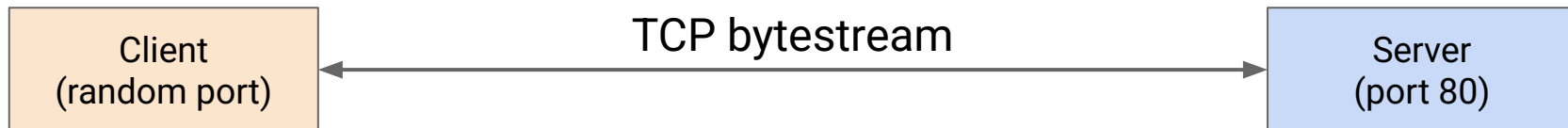
- One user is the client (e.g. your web browser, your terminal).
- One user is the server (e.g. the website).

HTTP **runs over TCP**.

- Client and server run a TCP handshake, and send data over the bytestream.
- No need to worry about packets being reordered, dropped, etc.
- Server listens for HTTP on well-known port 80. (*A later secure version uses 443.*)

HTTP is a **request-response** protocol.

- Client sends one request, and receives exactly one response.



HTTP Requests

The request syntax is in human-readable plaintext (can be typed by a human).

Version: What HTTP version we're using.

URL: The *resource* we want to interact with.

- Intuition: The filepath of a file on some remote server.

Method: What we want to do with that resource.

- GET: Send me this resource. Originally, this was the only method.
- POST: Send data to the server (e.g. user submits a form).
- Other methods for *manipulating* content on the server, not just retrieving it:
 - PUT, CONNECT, DELETE, OPTIONS, PATCH, TRACE, etc.

GET	/projects/project1.html	HTTP/1.1	\r\n
Method	URL	version	ends with a newline

HTTP Responses

Version: What HTTP version we're using.

Status code: A number, telling us what happened with the request.

Description: A description of the status code.

Content: The resource the user requested!

HTTP/1.1

version

200

status
code

OK

description

<html>Project 1 Spec...</html>

content

Status codes are used by the server to propagate information about the result of the request to the client.

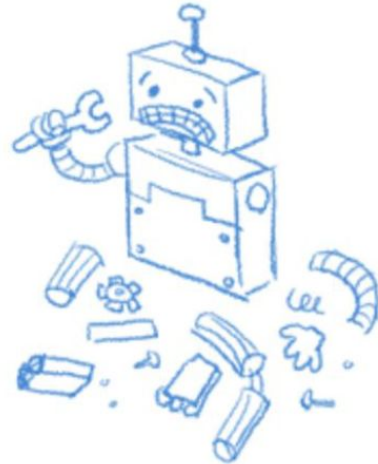
Codes are classified into various categories, according to numeric value.

- 100s: Informational responses.
- 200s: Successful responses.
- 300s: Redirection messages.
- 400s: Client error.
- 500s: Server error.



404. That's an error.

The requested URL /doesnotexist was not found on this server. That's all we know.



200s: Successful responses.

- **200 OK**: Request was successful.
 - Definition of success depends on the method in the request (e.g. GET, POST).
- **201 Created**: Request succeeded, and some new resource was created.
 - Seen generally in POST or PUT requests.



The [HTTP status dog](#) for
203 Non-Authoritative Information.

300s: Redirection messages.

- Used when a server is telling a client they should go and look for the resource (specified by the URL) somewhere else.
- **301 Moved Permanently.**
- **302 Found:** Moved temporarily.
- Response contains extra context about where the resource moved.
 - Location: `https://some.other.site/newpage.html`
 - Encoded in a header (more on this soon).

Status codes let the client determine future behavior.

- Example: 301 means, always go to the new location.
- Example: 302 means, come back here to check again in the future.

400s: Client error responses.

- **401 Unauthorized**: You need to authenticate (e.g. log in) to access this content.
- **403 Forbidden**: You are authenticated (server knows your identity), but access is still forbidden.
- **404 File Not Found**: You are requesting a file that doesn't exist.

500s: Server error responses.

- **500 Internal Server Error**: Server hit an error processing your request.
- **503 Service Unavailable**: Server cannot respond at the current time.

Status codes let the client determine future behavior.

- Example: 401 means, ask the user to log in.
- Example: 403 means, show an error message.

Sometimes, which status code we should use is ambiguous.

Example: Request the Google homepage with HTTP/0.9.

- Maybe Google should respond with: **505 HTTP Version Not Supported.**
- But Google actually responds with: **400 Bad Request.**

Usually, the category of error is the most important.

- In the example: 400 or 500 = error.
- Goal is to elicit the correct behavior from the client.

Requests and responses can contain additional metadata in the form of **headers**.

- Headers aren't mandatory (though server/client might expect a header and error).

Some headers are optional information.

- **User-Agent**: What program (e.g. Firefox, Chrome) the client is using.
- *Could* result in different processing of the request.

Some headers are critical information.

- **Content-Type**: File type of the response. (e.g. HTML, JPEG image, MP4 video...)

Headers can be classified into three types.

Request headers pass information about the client to the server.

- **Accept**: What file type the client is expecting in the response. Examples:
 - `Accept: text/html`
 - `Accept: application/json`
 - `Accept: image/*`
- **Host**: If a server is hosting multiple websites, identifies which website the client is aiming to access.
 - `Host: google.com:80`
- **Referer**: How the client triggered this request (e.g. clicking a link on Facebook).
- **User-Agent**: What program (e.g. Firefox, Chrome) the client is using.

"Referer" was [misspelled](#) in the original spec. Oops.

Response headers are in the response, but *not* directly related to the content.

- **Date**: When the server generated the response.
- **Location**: In 300 redirect responses, where the content moved to.
- **Server**: What software the server used to generate the response.

Representation headers are used in both requests and responses to describe how the content is represented.

- **Content-Type**: File type of the response. (e.g. HTML, JPEG image, MP4 video...)
- **Content-Encoding**: How the response is encoded into bits.
 - **Content-Encoding: gzip** says the contents were compressed.

HTTP Examples

Lecture X, CS 168, Fall 2024

HTTP

- Protocol Specification
- **Examples**
- Speeding Up HTTP

Content Delivery Networks

- Deployment
- Directing Clients to Caches

Newer HTTP Versions

HTTP in Terminal

```
$ telnet google.com 80
```



HTTP is a text-based protocol, so we can connect to the Google server, type requests, and read responses, all in the terminal.

```
Trying 2607:f8b0:4005:802::200e...
```

```
Connected to google.com.
```

```
Escape character is '^]'.
```

Using port 80 for HTTP.

```
GET / HTTP/1.1
```



Request: Get homepage, using HTTP version 1.1.

```
User-Agent: robjs
```



Adding a header to tell the server what type of client I'm using.

HTTP in Terminal

```
$ telnet google.com 80
```

```
Trying 2607:f8b0:4005:802::200e...
```

```
Connected to google.com.
```

```
Escape character is '^]'.
```

```
GET / HTTP/1.1
```

```
User-Agent: robjs
```

```
HTTP/1.1 200 OK
```

```
Date: Sat, 16 Mar 2024 18:33:08 GMT
```

```
Content-Type: text/html; charset=ISO-8859-1
```

```
<!doctype html><html lang="en"><head><meta content="Search the  
world's information, including webpages, images, videos and more.  
Google has many special features to help you find exactly what you're  
looking for." name="description">...
```

Response starts with status code: 200 OK.

Headers tell us the response date, file type (HTML), and encoding (e.g. ASCII).

The page we requested. (Would look nicer in a browser.)

HTTP Examples

Note: We need a URL in all requests (even POST requests).

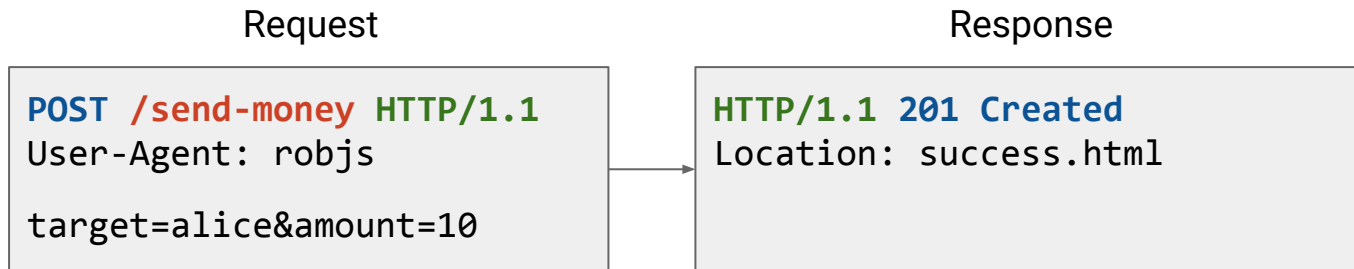
- The URL tells the server how to parse the information in the body of the request.
- Example: **POST** **/send-money** and **POST** **/request-money** do different things.

Note: The HTTP request can contain data.

- POST and PUT requests might contain data.
- GET requests probably don't contain data.

Note: This response doesn't have any content.

- The Location header redirects the user to the `success.html` page.

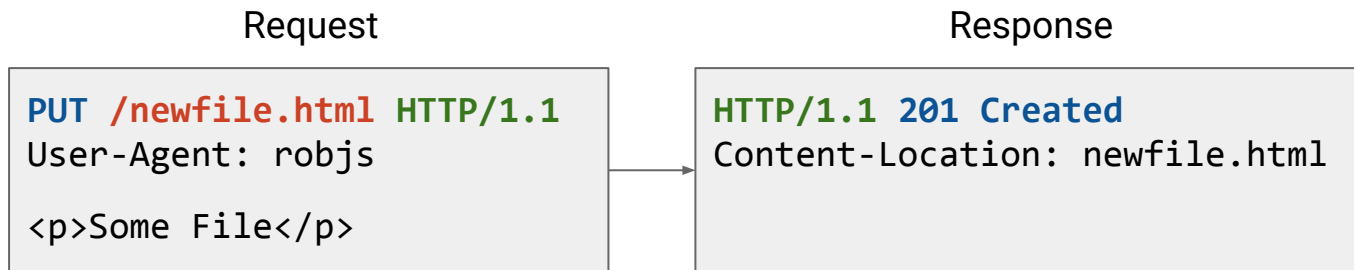


HTTP Examples

The request has data (PUT a file on the server).

The response does not have data.

The Content-Location header says that the file we uploaded is stored at newfile.html.



Speeding Up HTTP

Lecture X, CS 168, Fall 2024

HTTP

- Protocol Specification
- Examples
- **Speeding Up HTTP**

Content Delivery Networks

- Deployment
- Directing Clients to Caches

Newer HTTP Versions

Multiple HTTP Requests

Loading a single website can require multiple HTTP requests.

- One request for the HTML (text/formatting) of the page.
- Separate requests for every picture.
- Separate requests for scripts to make the page interactive.

Naive approach: Separate TCP connection for each request.

- We have to do a 3-way handshake for every request.



Multiple HTTP Requests – Pipelining

Smarter approach: Allow multiple requests to be **pipelined** over the same TCP connection.

- Trade-off: The server must maintain more open connections.



HTTP Cache Types

Optimization: Cache data to avoid sending duplicate requests for the same content.

Naive approach: Every request goes to to the origin server.

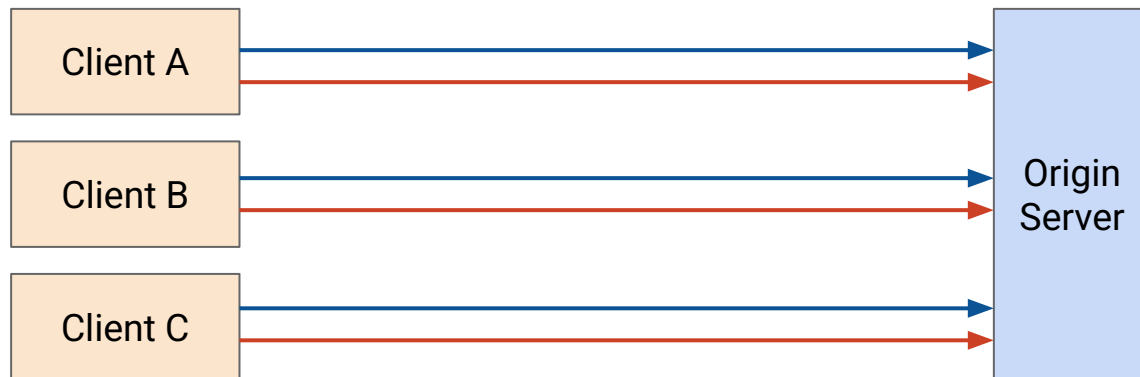
- Origin server: The server with the true (not cached) version of the content.

There are 3 types of caches: Private, Proxy, Managed.

In this diagram, 3 clients each request the same resource twice.

→ = first request

→ = second request



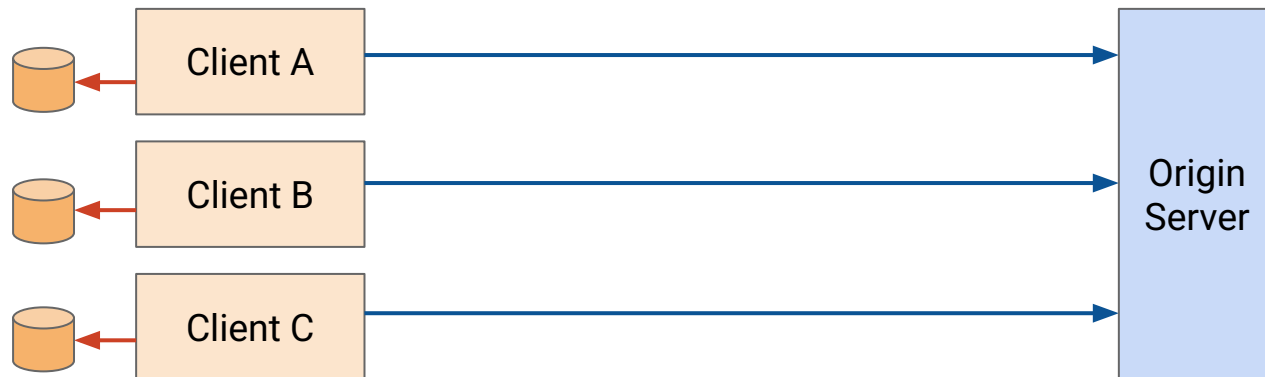
Private caches are tied to a specific end client (e.g. in a user's browser).

- When a user requests something for the second time, they can use the cache.

In this diagram, 3 clients each request the same resource twice.

→ = first request

→ = second request



Proxy caches are in the network (not end host).

- The first client's first request goes to the origin server.
- All subsequent requests can be served by the proxy cache.

Proxy caches are operated by a third party (not the client or server).

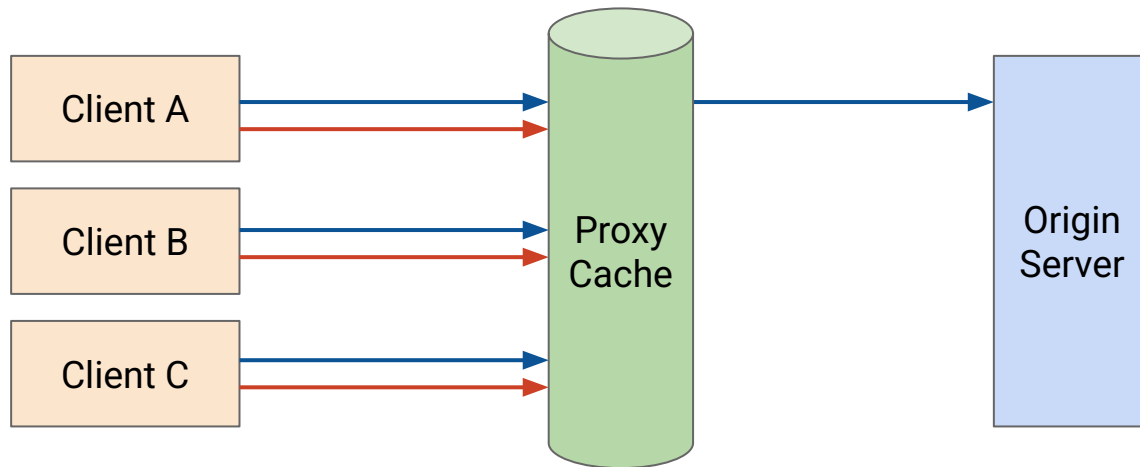
Problem: Clients need to be redirected to the proxy cache somehow.

- DNS resolver could lie and say, "server's IP address is [*proxy cache address*]."

In this diagram, 3 clients each request the same resource twice.

→ = first request

→ = second request



Managed caches are in the network (just like proxy caches).

Managed caches are operated by the server (but cache server $\neq$ origin server).

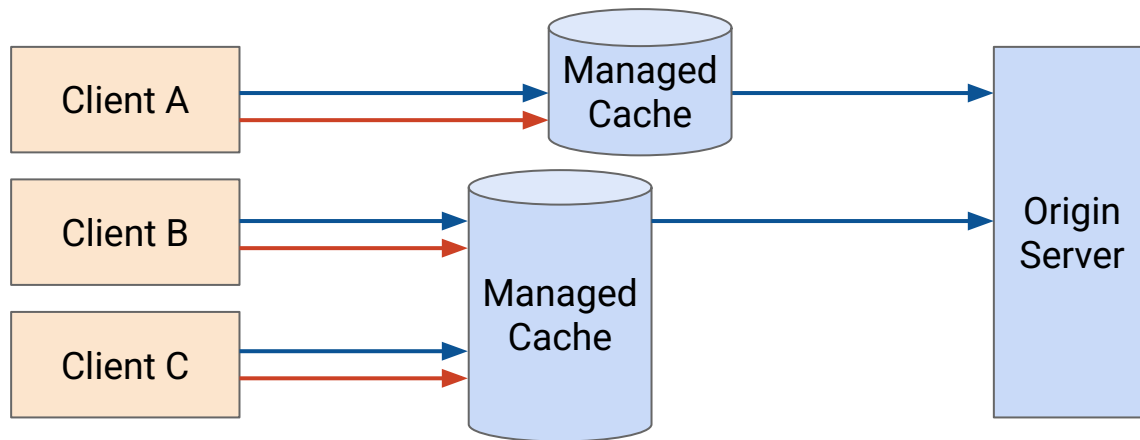
- Gives the server more control. Proxy cache might serve outdated/incorrect data.
- Server can redirect client to cache, because server knows about the caches.
 - In the HTML page: "Load the logo image from cache.google.com."

There can be multiple proxy/managed caches in the network.

In this diagram, 3 clients each request the same resource twice.

→ = first request

→ = second request



Implementing Caching – Static vs. Dynamic Content

HTTP resources can be static or dynamic.

- **Static:** Stays the same for a long time. (e.g. Google logo image.)
- **Dynamic:** Generated on-demand for every request. (e.g. Search results.)

The server needs to tell everybody whether data can be cached, and if so, for how long.

- We can use HTTP headers!
- **Expires** header tells us when the content can be cached until.
 - Used in HTTP/1.0, obsoleted in HTTP/1.1.

Servers can't enforce that clients and caches actually obey the header.

- The header is more of a *request* to cache than a contract.

```
HTTP/1.0 200 OK
```

```
Date: Sat, 16 Mar 2024 19:40:24 GMT
```

```
Expires: Sun, 17 Mar 2024 19:40:24 GMT
```

The data in this response
can be cached for 24 hours!

Implementing Caching – Cache-Control Header

The `Cache-Control` header lets the server give more details on how to cache the data.

- `Cache-Control: private, max-age=86400`
 - Store in private caches for 24 hours.
 - Useful for responses that are different per user (private cache only).
- `Cache-Control: no-store`
 - This content cannot be cached.
- More complex policies are possible.
 - Example: Use the HEAD request method to re-request the header only and re-validate the data before using cache.

```
HTTP/1.1 200 OK
```

```
Date: Sat, 16 Mar 2024 19:40:24 GMT
```

```
Expires: Sun, 16 Mar 2024 19:40:24 GMT
```

```
Cache-Control: private, max-age=31536000
```

Servers could include both Expires (1.0) and Cache-Control (1.1) headers for compatibility.

1.1 client might ignore 1.0 header (and vice-versa).

Benefits of Caching

Caching benefits everybody.

- Client can load pages faster.
 - With private caches, client saves time on subsequent accesses.
 - With proxy/managed caches, client gets data from a closer source.
 - Closer sources = lower RTT = higher TCP throughput.
- Less network bandwidth is needed.
 - Proxy caches are useful when there's low bandwidth out of a network.
- Reduces load on origin server.

Conveniently, the larger objects are *static*.

- Go to the origin server for dynamic content (e.g. small HTML page).
- Use a cache for static content (e.g. images, videos).

Content Delivery Networks

Lecture X, CS 168, Fall 2024

HTTP

- Protocol Specification
- Examples
- Speeding Up HTTP

Content Delivery Networks

- **Deployment**
- Directing Clients to Caches

Newer HTTP Versions

How can application providers use caching to improve load time for users?

- Private caches are only useful if the same user accesses the same content repeatedly.
- Proxy caches aren't managed by the server.
 - Need some changes to tell the client about the caches (e.g. DNS trick).
 - Proxy cache might not obey the rules in the header.
- Managed caches are the best choice.
 - Controlled by the application provider.
 - Can be placed "close" to end users.
 - Application provider can redirect users to the caches.

Content Delivery Networks (CDNs): Deployments of servers that can serve content (HTTP resources).

Servers can be placed "close" to end users.

- Geographically.
- From a network perspective. (e.g. number of hops)

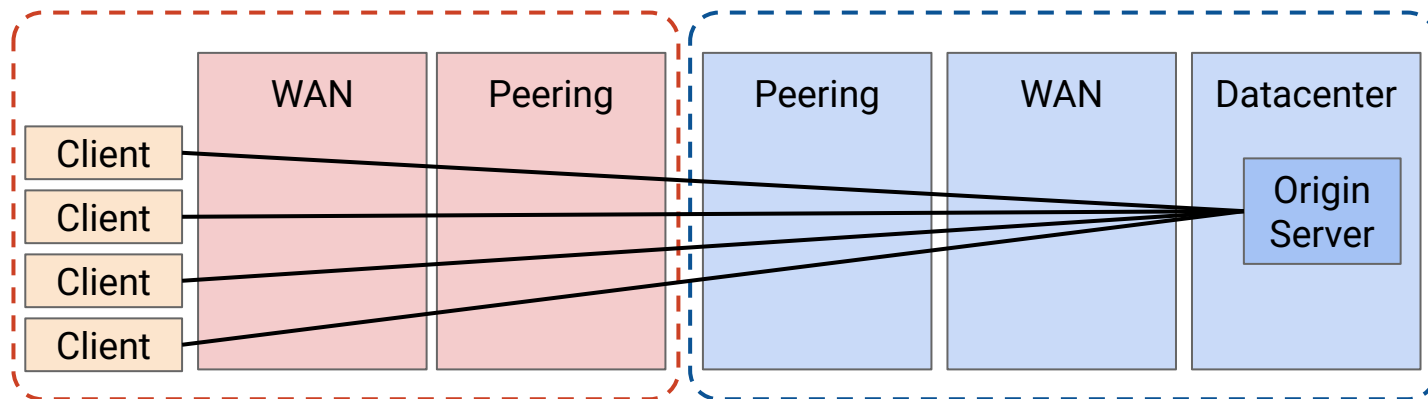
Benefits of CDNs:

- Higher performance for users.
 - Low-latency, high-throughput access to nearby server.
- Significant reductions in the bandwidth needed in the network.
- Reduces scaling needed for server infrastructure.
 - Adding more servers is easier than building one huge server.
- Provides better redundancy. If a server goes down, use another one.

Deploying CDNs

Without CDNs, every request reaches the origin server.

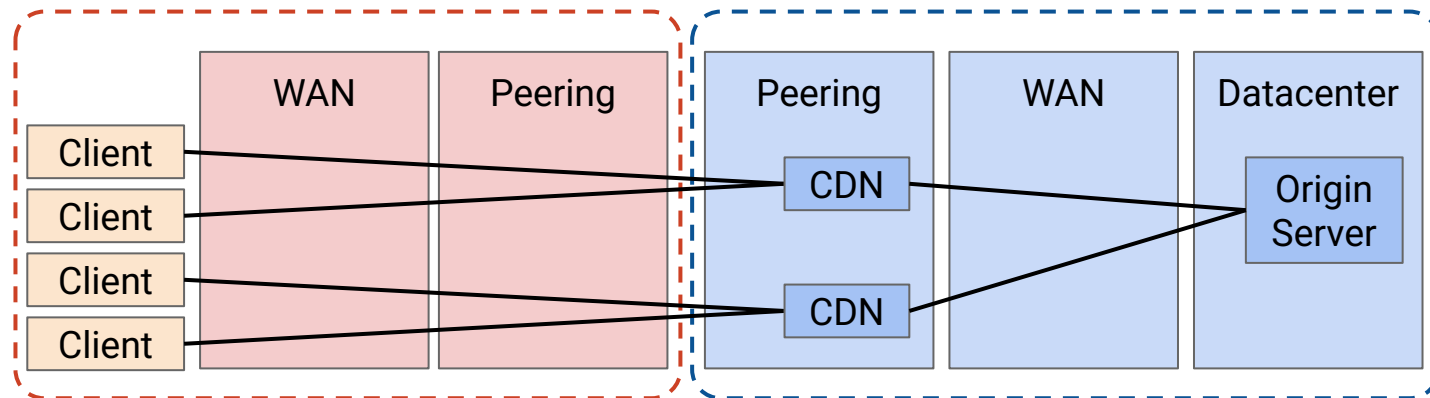
- Maximum latency → lowest performance.
- Maximum amount of "backbone" network traversed → highest cost to build.
- Scale must be supported on the origin server.



Deploying CDNs

Application provider could deploy CDNs in its own network.

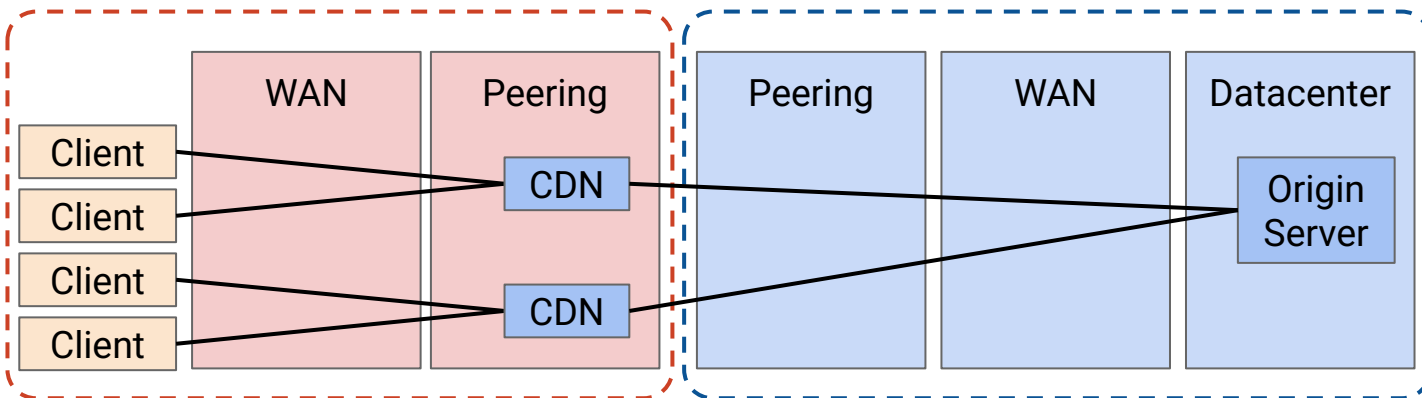
- Smaller servers at the "edge" of the application provider's networks.
- Reduces the volume of backbone traffic for the application provider.
- Reduces scale per deployment.



Deploying CDNs

Caching can be pushed "deeper" into the network.

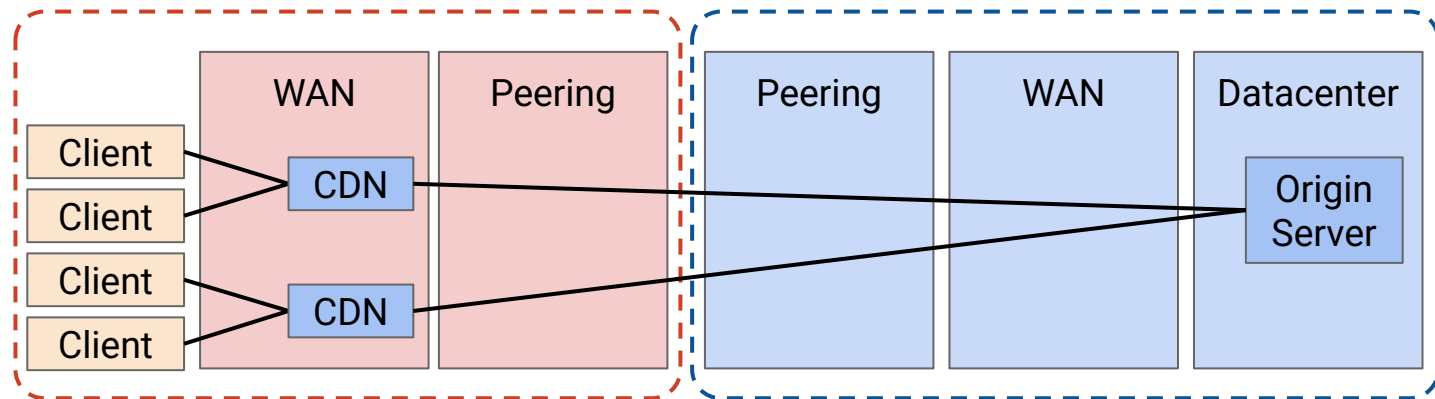
- Deploying CDNs in the ISP's network improves performance and reduces cost.
- ISP reduces their backbone network cost.
 - Less peering infrastructure needed.



Deploying CDNs

Deployment depth is limited by efficiency.

- Need many users accessing the same content to justify the cost of the CDN.
- Cost-benefit trade-off between:
 - Cost of building a CDN.
 - Cost savings from building less network capacity.
- Example: Deploying a CDN in every user's home is probably not worth the cost.



Large Global CDNs

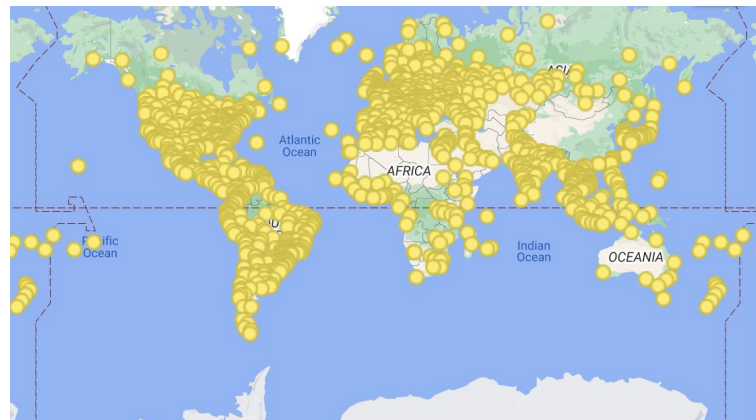
Large application providers host their own CDNs.

- Netflix, Google, Amazon, Facebook.

CDN providers can host your service on their infrastructure for a fee.

- Cloudflare, Akamai, Edgio.
- Useful for smaller application providers.

Deployments either in their own networks, or directly into ISP networks.



ISP companies often have their own content to serve.

- Example: Video-on-demand, or live TV, as part of TV+Internet bundles.
- CDN server infrastructure is also deployed by these ISPs.

Often a need for both third-party caches and the ISP's own infrastructure.

- [Sandvine report \(2023\):](#)
 - 15% of all Internet traffic is Netflix.
 - 11.4% is YouTube.
 - 4.5% is Disney+.
- Deploying caches can mean reducing ~25% of network capacity!

CDN servers are highly optimized for content delivery and storage.

Flash appliance focus areas

- 2U for rack efficiency (no deeper than 29 inches)
- Enough low cost NAND to reach 24GB/s of throughput (<0.3 DWPD)
- Connect at up to 2X100G LAG
- 2 and 4 post racking
- AC or DC power
- Single processor

Storage appliance focus areas

- Large storage capacity
- 2U for rack efficiency (no deeper than 29 inches)
- Enough low cost NAND to reach 10GB/s of throughput (<0.3 DWPD)
- Network flexibility to connect at 6x100 LAG or up to 2x100GE
- 2 and 4 post racking
- AC or DC power
- Single processor

Example of Netflix server specs (you don't need to understand these).

CDNs are mutually beneficial!

- Content provider gets better performance.
 - Better performance = more customers for both application and ISP.
- ISP gets lower bandwidth costs.

Cooperative commercial model:

- Content provider provides the servers for free.
- ISP hosts the servers for free.

In some cases, commercial negotiations are required.

- ISP and provider costs might not be equal as we get "deeper" into the network.

Becomes more difficult as there are more caching providers.

Commercial Challenges – Fragmentation

Cache deployment makes sense if there are small numbers of large content providers.

There's a long tail of smaller content providers.

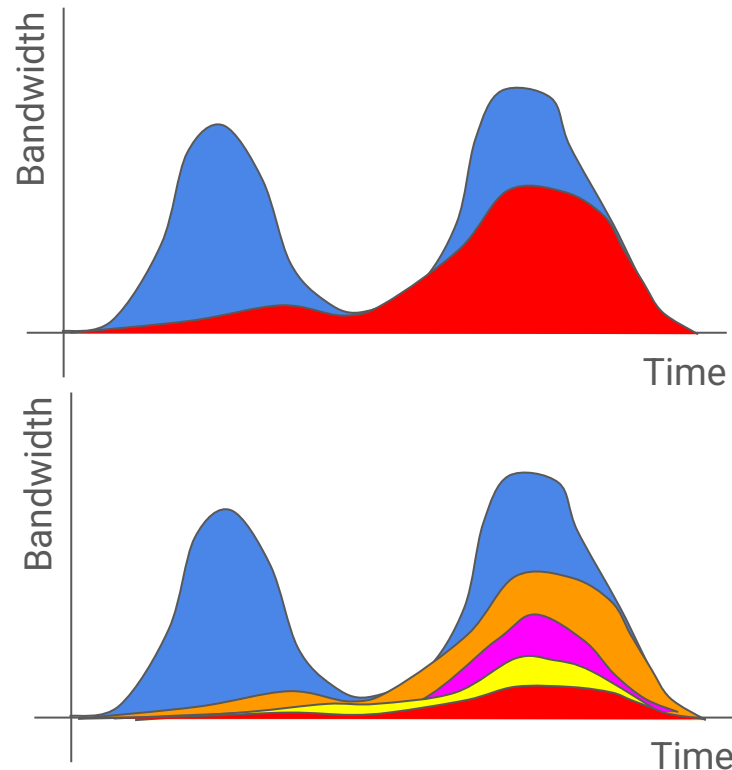
- Sandvine (2023): 4.5% is Disney+, 2.8% is Amazon Prime.

Idea: Can we have shared caching infrastructure?

- CDN InterConnect ([CDNI](#)) – IETF
- [OpenCaching](#)

Shared infrastructure is challenging!

- Who ensures quality?
- How are resources shared?



Directing Clients to Caches

Lecture X, CS 168, Fall 2024

HTTP

- Protocol Specification
- Examples
- Speeding Up HTTP

Content Delivery Networks

- Deployment
- **Directing Clients to Caches**

Newer HTTP Versions

Recall our CDN model:

- Go to the origin server for dynamic content (e.g. small HTML page).
- Origin server directs the user to a cache for static content (e.g. images, videos).

If there are many CDN servers, which one should the server direct the user to?

Three approaches:

- Anycast.
- DNS-based load balancing.
- Application-level mapping.

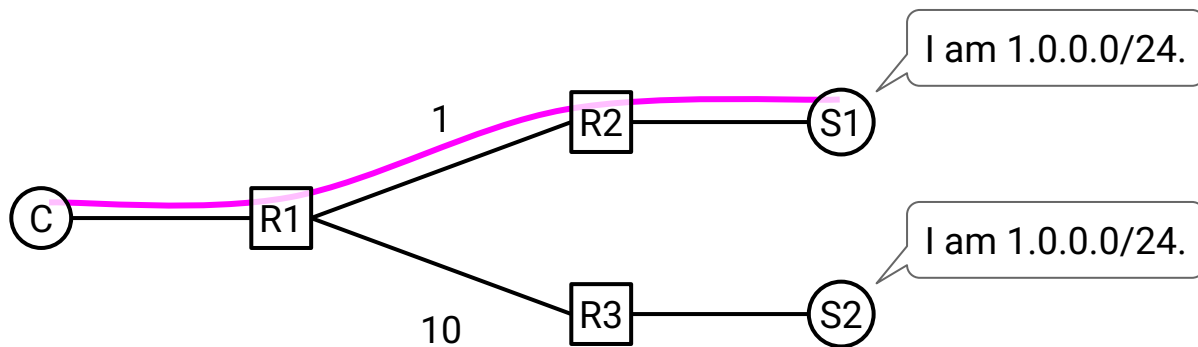
Directing Clients to Caches – Anycast

Recall **anycast**: Advertise the same IP prefix from multiple locations.

- Least-cost routing chooses the best server to use.

Problem: Routing can change in the middle of a long-lived connection.

- Client starts a TCP connection with S1.
- Network topology changes. Client's packets are sent to S2 now.
- Server B doesn't have a TCP connection open and gets confused.
- Key issue: Routing thinks S1 and S2 are the same (since they have the same IP.)



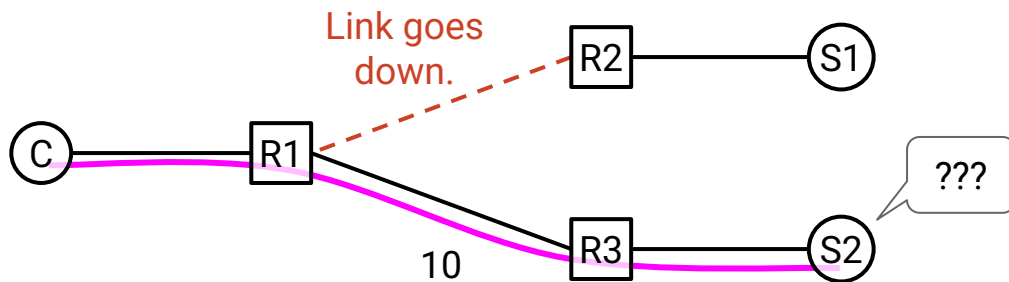
Directing Clients to Caches – Anycast

Recall anycast: Advertise the same IP prefix from multiple locations.

- Least-cost routing chooses the best server to use.

Problem: Routing can change in the middle of a long-lived connection.

- Client starts a TCP connection with S1.
- Network topology changes. Client's packets are sent to S2 now.
- Server B doesn't have a TCP connection open and gets confused.
- Key issue: Routing thinks S1 and S2 are the same (since they have the same IP).



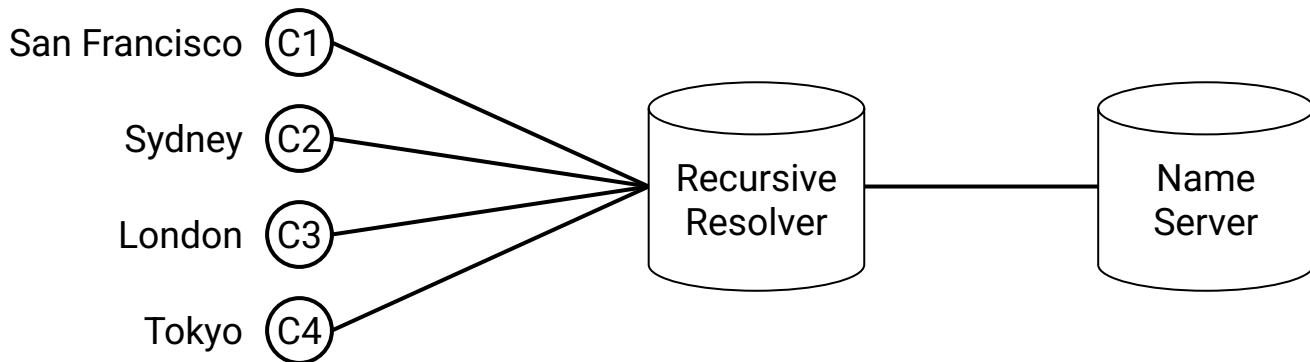
Directing Clients to Caches – DNS-Based Load Balancing

Recall DNS-based load balancing: Map the same domain to different IP addresses, depending on where the query came from.

- Solves the anycast problem: Different servers have different addresses.

Problem: Granularity.

- If many users query through the same resolver, they all get the same address.
- Extensions are required to add end user information in the query.



Directing Clients to Caches – Application-Level Mapping

Application directs the user to the cache.

- In the HTML page: "Load the video from sfo-cache.google.com."
- In the HTML page: "Load the video from mia-cache.google.com."

Benefits:

- Application knows the end user's address. (Solves the DNS granularity problem.)
- Allows for mapping at per-content item granularity.
 - Popular cat videos are served from many servers.
 - Obscure videos are served from fewer servers, closer to the origin.

Drawbacks:

- Still need to figure out which cache is "closest" to the client.
- Still need to decide the right strategy for failures (e.g. if a server goes down).

Newer HTTP Versions

Lecture X, CS 168, Fall 2024

HTTP

- Protocol Specification
- Examples
- Speeding Up HTTP

Content Delivery Networks

- Deployment
- Directing Clients to Caches

Newer HTTP Versions

Lots of applications run over HTTP.

- HTTP is very flexible. It can serve videos, webpages, etc.

As HTTP got popular, security became a concern.

- Network attackers (e.g. a malicious router) can read HTTP content.

HTTPS is a secure version of the protocol.

- Same syntax and semantics, but runs over TLS-over-TCP, instead of just TCP.
- TLS: Client and server exchange secret keys and encrypt their messages.
- Majority of traffic on the Internet (85.4% of websites) are now default HTTPS.
 - 85.4% of websites, according to [W3Techs](#).
- HTTPS listens over port 443 (instead of 80).

HTTP/2.0 was introduced in 2015. (First new revision since 1997!)

Aims to decrease latency and improve page load speed.

- Data compression of headers.
- Server-side pushing: Servers can preemptively send data, without waiting for a request.
 - HTTP/2.0 is no longer a strict request-response protocol.
- Prioritization of requests.
- Better multiplexing of simultaneous requests.
 - Example: Ensure a small response doesn't get stuck behind a large one.

Widely adopted across client software (e.g. browsers) and CDNs.

HTTP/3.0 was introduced in 2022. (Not long after the previous update!)

Semantics are the same as HTTP/2.0, but runs over QUIC instead of TCP.

- QUIC = Quick UDP Connections.
 - Designed at Google. Standardized in IETF.
- QUIC is custom-built to work well with HTTP.
 - Abandons a classic network paradigm (layering).
 - In exchange, we get to simultaneously optimize Layer 7 and Layer 4 together.

Summary: HTTP and CDNs

- HTTP is a protocol used to transfer data between a client and server.
 - Originally designed for HTML web pages.
- HTTP consists of request and response messages with headers in them.
 - Allows for different types of content to be carried over it.
- Performance of HTTP can be improved through caching static content.
 - HTTP provides means to control how this caching is used.
- Content Delivery Networks (CDNs) provide infrastructure to allow for this caching to be implemented to improve application performance.